VP08 — Channel contracts and sequences

Who communicates, what is exchanged, and in what order?

Explicit scenario steps validated against permits, participation, mode and request/result correlation.

Operating mode: All declared modes · MicroConsole · MicroCore · MicroRetained

Viewpoint overview

Contracts

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Each chapter uses the same model content as its object detail. Definitions without relationships remain visible. Large diagram sets are deferred to object details; all chapter statements and object links remain present.

CandidateLocalAddressContract

contract declaration: CandidateLocalAddressContract — Contracts/MicroValues.design, line 5.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: CandidateLocalAddress.

CandidateLocalAddressByte fields

Field	Value type	Presence	Source statement
GroupNibble	unsigned	required	CandidateLocalAddressByte has-field GroupNibble. — Contracts/MicroValues.design, line 38
MemberNibble	unsigned	required	CandidateLocalAddressByte has-field MemberNibble. — Contracts/MicroValues.design, line 39

Declared variants: CandidateLocalAddressByte.

Explicit properties

Property	Value	Source statement
completeness	closed	CandidateLocalAddressContract has completeness = closed. — Contracts/MicroValues.design, line 41

Declared relationships

- CandidateLocalAddress upholds CandidateLocalAddressContract. — Contracts/MicroValues.design, line 37
- CandidateLocalAddressByte has-field GroupNibble. — Contracts/MicroValues.design, line 38
- CandidateLocalAddressByte has-field MemberNibble. — Contracts/MicroValues.design, line 39

- CandidateLocalAddressContract defines CandidateLocalAddressByte. — Contracts/MicroValues.design, line 40

Related objects in this viewpoint

CandidateLocalAddress, CandidateLocalAddressByte, GroupNibble, MemberNibble

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

CandidatePresentationContract

contract declaration: CandidatePresentationContract — Contracts/MicroValues.design, line 9.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: CandidatePresentation.

CandidatePresentationByte fields

Field	Value type	Presence	Source statement
DecimalPlaces	unsigned	required	CandidatePresentationByte has-field DecimalPlaces. — Contracts/MicroValues.design, line 49
UnitCode	unsigned	required	CandidatePresentationByte has-field UnitCode. — Contracts/MicroValues.design, line 50

Declared variants: CandidatePresentationByte.

Explicit properties

Property	Value	Source statement
completeness	closed	CandidatePresentationContract has completeness = closed. — Contracts/MicroValues.design, line 52

Declared relationships

- CandidatePresentation upholds CandidatePresentationContract. — Contracts/MicroValues.design, line 48
- CandidatePresentationByte has-field DecimalPlaces. — Contracts/MicroValues.design, line 49
- CandidatePresentationByte has-field UnitCode. — Contracts/MicroValues.design, line 50
- CandidatePresentationContract defines CandidatePresentationByte. — Contracts/MicroValues.design, line 51

Related objects in this viewpoint

CandidatePresentation, CandidatePresentationByte, DecimalPlaces, UnitCode

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

CommittedRecordsShape

contract declaration: CommittedRecordsShape — Contracts/MicroValues.design, line 14.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: CommittedRecords.

CommittedRecordsShape fields

Field	Value type	Presence	Source statement
CommittedGeneration	unsigned	required	CommittedRecordsShape has-field CommittedGeneration. — Contracts/MicroValues.design, line 64
CommittedPayload	bytes	required	CommittedRecordsShape has-field CommittedPayload. — Contracts/MicroValues.design, line 65
IntegrityMarker	unsigned	required	CommittedRecordsShape has-field IntegrityMarker. — Contracts/MicroValues.design, line 66

Explicit properties

Property	Value	Source statement
completeness	closed	CommittedRecordsShape has completeness = closed. — Contracts/MicroValues.design, line 63

Declared relationships

- CommittedRecords upholds CommittedRecordsShape. — Contracts/MicroValues.design, line 62
- CommittedRecordsShape has-field CommittedGeneration. — Contracts/MicroValues.design, line 64
- CommittedRecordsShape has-field CommittedPayload. — Contracts/MicroValues.design, line 65
- CommittedRecordsShape has-field IntegrityMarker. — Contracts/MicroValues.design, line 66

Related objects in this viewpoint

CommittedGeneration, CommittedPayload, CommittedRecords, IntegrityMarker

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

NativeTaskApiContract

contract declaration: NativeTaskApiContract — Contracts/MicroServices.design, line 8.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: NativeTaskApi.

Allowed payloads: TaskInvocation, TaskReturn. Request/result correlation requires explicit replies-to relationships and scenario steps.

No record fields are declared for this contract or its variants. A channel contract may describe permitted messages without defining record fields.

Explicit properties

Property	Value	Source statement
completeness	closed	NativeTaskApiContract has completeness = closed. — Contracts/MicroServices.design, line 54

Declared relationships

- NativeTaskApi upholds NativeTaskApiContract. — Contracts/MicroServices.design, line 53
- NativeTaskApiContract permits TaskInvocation. — Contracts/MicroServices.design, line 55

- NativeTaskApiContract permits TaskReturn. — Contracts/MicroServices.design, line 56

Related objects in this viewpoint

NativeTaskApi, TaskInvocation, TaskReturn

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

NumericCommandApiContract

contract declaration: NumericCommandApiContract — Contracts/MicroServices.design, line 10.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: NumericCommandApi.

Allowed payloads: CommandRequest, CommandResult. Request/result correlation requires explicit replies-to relationships and scenario steps.

No record fields are declared for this contract or its variants. A channel contract may describe permitted messages without defining record fields.

Explicit properties

Property	Value	Source statement
completeness	closed	NumericCommandApiContract has completeness = closed. — Contracts/MicroServices.design, line 58

Declared relationships

- NumericCommandApi upholds NumericCommandApiContract. — Contracts/MicroServices.design, line 57

- NumericCommandApiContract permits CommandRequest. — Contracts/MicroServices.design, line 59
- NumericCommandApiContract permits CommandResult. — Contracts/MicroServices.design, line 60

Related objects in this viewpoint

CommandRequest, CommandResult, NumericCommandApi

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

NumericRequestShape

contract declaration: NumericRequestShape — Contracts/MicroServices.design, line 11.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: CommandRequest, ReadValueRequest, WriteValueRequest.

NumericRequestShape fields

Field	Value type	Presence	Source statement
RequestMember	unsigned	required	NumericRequestShape has-field RequestMember. — Contracts/MicroServices.design, line 62
RequestOwner	unsigned	required	NumericRequestShape has-field RequestOwner. — Contracts/MicroServices.design, line 63

RequestPayload	bytes	optional	NumericRequestShape has-field RequestPayload. — Contracts/MicroServices.design, line 64

Explicit properties

Property	Value	Source statement
completeness	closed	NumericRequestShape has completeness = closed. — Contracts/MicroServices.design, line 61

Declared relationships

- CommandRequest upholds NumericRequestShape. — Contracts/MicroServices.design, line 42
- NumericRequestShape has-field RequestMember. — Contracts/MicroServices.design, line 62
- NumericRequestShape has-field RequestOwner. — Contracts/MicroServices.design, line 63
- NumericRequestShape has-field RequestPayload. — Contracts/MicroServices.design, line 64
- ReadValueRequest upholds NumericRequestShape. — Contracts/MicroServices.design, line 75
- WriteValueRequest upholds NumericRequestShape. — Contracts/MicroServices.design, line 128

Related objects in this viewpoint

CommandRequest, ReadValueRequest, RequestMember, RequestOwner, RequestPayload, WriteValueRequest

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

NumericValueApiContract

contract declaration: NumericValueApiContract — Contracts/MicroServices.design, line 13.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: NumericValueApi.

Allowed payloads: ReadValueRequest, ReadValueResult, WriteValueRequest, WriteValueResult. Request/result correlation requires explicit replies-to relationships and scenario steps.

No record fields are declared for this contract or its variants. A channel contract may describe permitted messages without defining record fields.

Explicit properties

Property	Value	Source statement
completeness	closed	NumericValueApiContract has completeness = closed. — Contracts/MicroServices.design, line 66

Declared relationships

- NumericValueApi upholds NumericValueApiContract. — Contracts/MicroServices.design, line 65
- NumericValueApiContract permits ReadValueRequest. — Contracts/MicroServices.design, line 67
- NumericValueApiContract permits ReadValueResult. — Contracts/MicroServices.design, line 68
- NumericValueApiContract permits WriteValueRequest. — Contracts/MicroServices.design, line 69
- NumericValueApiContract permits WriteValueResult. — Contracts/MicroServices.design, line 70

Related objects in this viewpoint

NumericValueApi, ReadValueRequest, ReadValueResult, WriteValueRequest, WriteValueResult

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

OperationResultShape

contract declaration: OperationResultShape — Contracts/MicroServices.design, line 14.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: CommandResult, CommitResult, ReadValueResult, RecoverResult, TaskReturn, WriteValueResult.

OperationResultShape fields

Field	Value type	Presence	Source statement
ResultPayload	bytes	optional	OperationResultShape has-field ResultPayload. — Contracts/MicroServices.design, line 72
ResultStatus	unsigned	required	OperationResultShape has-field ResultStatus. — Contracts/MicroServices.design, line 73

Explicit properties

Property	Value	Source statement
completeness	closed	OperationResultShape has completeness = closed. — Contracts/MicroServices.design, line 71

Declared relationships

- CommandResult upholds OperationResultShape. — Contracts/MicroServices.design, line 45
- CommitResult upholds OperationResultShape. — Contracts/MicroServices.design, line 50
- OperationResultShape has-field ResultPayload. — Contracts/MicroServices.design, line 72
- OperationResultShape has-field ResultStatus. — Contracts/MicroServices.design, line 73
- ReadValueResult upholds OperationResultShape. — Contracts/MicroServices.design, line 78
- RecoverResult upholds OperationResultShape. — Contracts/MicroServices.design, line 85
- TaskReturn upholds OperationResultShape. — Contracts/MicroServices.design, line 122
- WriteValueResult upholds OperationResultShape. — Contracts/MicroServices.design, line 131

Related objects in this viewpoint

CommandResult, CommitResult, ReadValueResult, RecoverResult, ResultPayload, ResultStatus, TaskReturn, WriteValueResult

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

RegistryBindingsShape

contract declaration: RegistryBindingsShape — Contracts/MicroValues.design, line 26.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: RegistryBindings.

RegistryBindingsShape fields

Field	Value type	Presence	Source statement
BindingOwnerId	unsigned	required	RegistryBindingsShape has-field BindingOwnerId. — Contracts/MicroValues.design, line 89
DisplayName	text	optional	RegistryBindingsShape has-field DisplayName. — Contracts/MicroValues.design, line 90
ExternalAccessPolicy	unsigned	required	RegistryBindingsShape has-field ExternalAccessPolicy. — Contracts/MicroValues.design, line 91
PackedLocalId	unsigned	required	RegistryBindingsShape has-field PackedLocalId. — Contracts/MicroValues.design, line 92
PresentationByte	unsigned	optional	RegistryBindingsShape has-field PresentationByte. — Contracts/MicroValues.design, line 93
ScalarKind	unsigned	required	RegistryBindingsShape has-field ScalarKind. — Contracts/MicroValues.design, line 94
ScalarReference	bytes	required	RegistryBindingsShape has-field ScalarReference. — Contracts/MicroValues.design, line 95

Explicit properties

Property	Value	Source statement
completeness	closed	RegistryBindingsShape has completeness = closed. — Contracts/MicroValues.design, line 88

Declared relationships

- RegistryBindings upholds RegistryBindingsShape. — Contracts/MicroValues.design, line 87
- RegistryBindingsShape has-field BindingOwnerId. — Contracts/MicroValues.design, line 89
- RegistryBindingsShape has-field DisplayName. — Contracts/MicroValues.design, line 90
- RegistryBindingsShape has-field ExternalAccessPolicy. — Contracts/MicroValues.design, line 91
- RegistryBindingsShape has-field PackedLocalId. — Contracts/MicroValues.design, line 92
- RegistryBindingsShape has-field PresentationByte. — Contracts/MicroValues.design, line 93
- RegistryBindingsShape has-field ScalarKind. — Contracts/MicroValues.design, line 94
- RegistryBindingsShape has-field ScalarReference. — Contracts/MicroValues.design, line 95

Related objects in this viewpoint

BindingOwnerId, DisplayName, ExternalAccessPolicy, PackedLocalId, PresentationByte, RegistryBindings, ScalarKind, ScalarReference

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

RetainedApiContract

contract declaration: RetainedApiContract — Contracts/MicroServices.design, line 26.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: RetainedApi.

Allowed payloads: CommitRequest, CommitResult, RecoverRequest, RecoverResult. Request/result correlation requires explicit replies-to relationships and scenario steps.

No record fields are declared for this contract or its variants. A channel contract may describe permitted messages without defining record fields.

Explicit properties

Property	Value	Source statement
completeness	closed	RetainedApiContract has completeness = closed. — Contracts/MicroServices.design, line 97

Declared relationships

- RetainedApi upholds RetainedApiContract. — Contracts/MicroServices.design, line 96
- RetainedApiContract permits CommitRequest. — Contracts/MicroServices.design, line 98
- RetainedApiContract permits CommitResult. — Contracts/MicroServices.design, line 99
- RetainedApiContract permits RecoverRequest. — Contracts/MicroServices.design, line 100
- RetainedApiContract permits RecoverResult. — Contracts/MicroServices.design, line 101

Related objects in this viewpoint

CommitRequest, CommitResult, RecoverRequest, RecoverResult, RetainedApi

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

RetainedRequestShape

contract declaration: RetainedRequestShape — Contracts/MicroServices.design, line 27.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: CommitRequest, RecoverRequest.

RetainedRequestShape fields

Field	Value type	Presence	Source statement
RecordPayload	bytes	optional	RetainedRequestShape has-field RecordPayload. — Contracts/MicroServices.design, line 103

Explicit properties

Property	Value	Source statement
completeness	closed	RetainedRequestShape has completeness = closed. — Contracts/MicroServices.design, line 102

Declared relationships

- CommitRequest upholds RetainedRequestShape. — Contracts/MicroServices.design, line 47
- RecoverRequest upholds RetainedRequestShape. — Contracts/MicroServices.design, line 82
- RetainedRequestShape has-field RecordPayload. — Contracts/MicroServices.design, line 103

Related objects in this viewpoint

CommitRequest, RecordPayload, RecoverRequest

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

ScheduleStateShape

contract declaration: ScheduleStateShape — Contracts/MicroValues.design, line 31.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: ScheduleState.

ScheduleStateShape fields

Field	Value type	Presence	Source statement
CurrentPeriod	unsigned	required	ScheduleStateShape has-field CurrentPeriod. — Contracts/MicroValues.design, line 104

LastDispatch	unsigned	required	ScheduleStateShape has-field LastDispatch. — Contracts/MicroValues.design, line 105
TaskEnabled	boolean	required	ScheduleStateShape has-field TaskEnabled. — Contracts/MicroValues.design, line 106

Explicit properties

Property	Value	Source statement
completeness	closed	ScheduleStateShape has completeness = closed. — Contracts/MicroValues.design, line 103

Declared relationships

- ScheduleState upholds ScheduleStateShape. — Contracts/MicroValues.design, line 102
- ScheduleStateShape has-field CurrentPeriod. — Contracts/MicroValues.design, line 104
- ScheduleStateShape has-field LastDispatch. — Contracts/MicroValues.design, line 105
- ScheduleStateShape has-field TaskEnabled. — Contracts/MicroValues.design, line 106

Related objects in this viewpoint

CurrentPeriod, LastDispatch, ScheduleState, TaskEnabled

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

TaskInvocationShape

contract declaration: TaskInvocationShape — Contracts/MicroServices.design, line 29.

Completeness: **closed**. This is a declared shape constraint, not observed validation or delivery.

Used by: TaskInvocation.

TaskInvocationShape fields

Field	Value type	Presence	Source statement
DispatchTimestamp	unsigned	required	TaskInvocationShape has-field DispatchTimestamp. — Contracts/MicroServices.design, line 119

Explicit properties

Property	Value	Source statement
completeness	closed	TaskInvocationShape has completeness = closed. — Contracts/MicroServices.design, line 118

Declared relationships

- TaskInvocation upholds TaskInvocationShape. — Contracts/MicroServices.design, line 117
- TaskInvocationShape has-field DispatchTimestamp. — Contracts/MicroServices.design, line 119

Related objects in this viewpoint

DispatchTimestamp, TaskInvocation

Interpretation and unspecified behavior

Participation topology is static. Only numbered steps within one scenario establish an order. Contracts declare allowed payloads; participation does not prove delivery. No transport, precondition, or runtime scheduling is inferred.

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